Tajbia Hossain

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Research Interests

Software engineer and researcher with a solid academic foundation in machine learning and natural language processing, complemented by over four years of systems engineering experience at Samsung R&D. My research focuses on developing robust, data-efficient representation learning methods and deep neural architectures for complex text understanding tasks. In particular, I am interested in designing algorithmic frameworks to address extreme class imbalance, low-resource settings, and noisy real-world text distributions. I aim to pursue graduate research investigating parameter-efficient adaptation, robust semantic representations, and scalable inference for next-generation language technologies.

Professional Experience

Engineer II (Lead Software Engineer) (Promoted from Engineer I) **Feb 2022 – Present**
Samsung R&D Institute Bangladesh (SRBD) Dhaka, Bangladesh

- **Technical & Systems Engineering:**
  - Designed and developed the primary UI/UX and architected connectivity subsystems for native Galaxy Book Windows applications, including Camera Share (C#, WPF) and Storage Share (C#, WinUI 3).
  - Integrated the Samsung Multi-Connectivity Framework (MCF) SDK for multi-device discovery and connectivity and tuned Real-Time Transport Protocol (RTP) streaming pipelines for ultra-low latency transmission.
  - Built cross-process pipelines using secure Named Pipes and Windows AppService, addressing security vulnerabilities and IPC communication across background services.
  - Implemented direct Win32 API controllers for Bluetooth and Wi-Fi, coupled with dynamic power-state handlers to manage device sleep/wake transitions.
  - Engineered a Proof-of-Concept (PoC) BLE connectivity module interfacing Samsung's Smart Ring with Windows PC; processed and parsed real-time 3-axis accelerometer telemetry to build training datasets for gesture-based cursor and mouse control models.
  - Designed 100+ CPU/memory profiling test suites and resolved major static analysis issues (SAM/MCD) to ensure stability across deployment cycles.
- **Leadership, Innovation & Mentorship:**
  - Managed end-to-end delivery of Camera Share and Storage Share under strict release timelines.
  - Researched and co-authored multiple patent filings focused on energy-efficient BLE discovery, dynamic Wi-Fi/P2P transfer thresholds, and multi-device orientation sensing.
  - Served on the Innovation Task Force and as a reviewer for the Mobile Developer Conference (MDC) idea contest, critically evaluating novel proposals and Description of Invention (DOI) documents that progressed toward patent filings.
  - Mentored 3 junior software engineers and directly supervised 2 research interns on multi-device systems and patent disclosures.

Education

BSc in Computer Science and Engineering Dhaka, Bangladesh
Ahsanullah University of Science and Technology (AUST) **Apr 2016 – Jan 2021**

- **Cumulative GPA:** 3.449 / 4.00
- **Undergraduate Thesis:** Text Classification using Deep Learning on Imbalanced text Data
- **Thesis Supervisor:** Raqeebir Rab, Assistant Professor

Test Scores

IELTS Academic: Overall Band 8.0 (Listening 9.0, Reading 8.5, Writing 7.5, Speaking 7.5) **Oct 2026**

Publications

[1] **T. Hossain**, H. Z. Mauni, and R. Rab, "Reducing the Effect of Imbalance in Text Classification Using SVD and GloVe with Ensemble and Deep Learning," *Computing and Informatics*, vol. 41, no. 1, pp. 98–115, 2022. [Journal]

[2] H. Z. Mauni, **T. Hossain**, and R. Rab, "Classification of Underrepresented Text Data in an Imbalanced Dataset Using Deep Neural Network," in *Proc. IEEE Region 10 Symposium (TENSYMP)*, Dhaka, Bangladesh, 2020, pp. 997–1000. [IEEE Xplore]

Patents

[1] M. M. M. Khan, M. M. Haque, M. A. Hossain, F. Tanvir, A. K. Majumder, and **T. Hossain**, "Proposal for Reducing BLE Packet Loss in Low-Power Scanning Mode," Patent Application, filed Jul. 2026.

[2] M. M. M. Khan, M. M. Haque, F. Tanvir, A. K. Majumder, M. A. Hossain, and **T. Hossain**, "Energy-Efficient Bluetooth Low Energy (BLE) Device Discovery," Patent Application, filed Mar. 2026.

[3] J. Islam, S. M. Monowara, M. A. N. Akhand, A. R. Noor, **T. Hossain**, and M. M. M. Khan, "Secure Group Photo Sharing using Facial Verification at Receiving End," Patent Application, filed Dec. 2025.

[4] M. M. M. Khan, M. M. Haque, **T. Hossain**, and A. D. Datta, "Efficient File Transfer System by Dynamically Determining Cut-off threshold of File Size," Patent Application, filed Mar. 2025.

[5] M. M. M. Khan, M. A. Quaum, M. K. Akon, M. M. Islam, N. Naznin, and **T. Hossain**, "An electronic device and Method for controlling the electronic device thereof," Republic of Korea Patent Application KR20250070416A, filed Aug. 2023. [Google Patents]

[6] K. I. Tori, M. A. M. Ganim, M. M. M. Khan, S. I. Ahmad, and **T. Hossain**, "Electronic device for communicating with external electronic device and storage medium thereof," Republic of Korea Patent Application KR20250089233A, filed Jun. 2023. [Google Patents]

[7] M. M. M. Khan, M. Rahman, S. I. Ahmad, and **T. Hossain**, "Method for controlling external electronic device and electronic device for supporting the same," Republic of Korea Patent Application KR20250024357A, filed Mar. 2023. [Google Patents]

Awards & Recognition

Outstanding Female Contributor Award **2023**
Samsung R&D Institute Bangladesh

- Conferred in recognition of exemplary engineering impact and leadership within the organization.

Excellence Award (4-time recipient) **2023, 2024, 2025**
Samsung R&D Institute Bangladesh Apr–Jun 2023, Oct–Dec 2023, Apr–Jun 2024, Apr–Jun 2025

- Recognized for exceptional software development performance and technical contributions across consecutive evaluation cycles.

Inventor Award **Jul–Sep 2023**
Samsung R&D Institute Bangladesh

- Awarded for outstanding research and innovation contributions leading to novel patent disclosures.

The Daily Star High Achievers Award **2015**
The Daily Star, Bangladesh

- Received for academic distinction in Edexcel IGCSE examinations.

References

Raqeebir Rab
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